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Regulation of Protein Synthesis and Apoptosis in Lymphocytes of Parkinson Patients: The Effect of Dopaminergic Treatment

Stéphanie Pain[a, b] Julie Deguil[b] Jeremie Belin[c] Pauline Barraud[b] Stéphanie Ragot[b, e] Jean-Luc Houeto[b, d, e]

[a]INSERM U1084, Laboratoire de Neurosciences Expérimentales et Cliniques, LNEC, Université de Poitiers, Poitiers, France; [b]Faculté de Médecine-Pharmacie, Université de Poitiers, Poitiers, France; [c]Centre Expert Parkinson, Service de Neurologie, CHU de Tours, Tours, France; [d]Service de Neurologie, CHU Poitiers, Poitiers, France; [e]INSERM, CIC 1402, Poitiers, France

Keywords
Parkinson disease · Lymphocytes · Protein synthesis · Apoptosis · Dopaminergic treatment

Abstract
Background: Parkinson disease (PD) is a neurodegenerative disorder characterized by progressive degeneration of the dopaminergic neurons in the substantia nigra, presumably due to increased apoptosis. In previous studies, we showed altered expression of proteins involved in mammalian target of rapamycin (mTOR) antiapoptotic and double-stranded RNA-dependent protein kinase (PKR) apoptotic pathways of translational control in experimental cellular and animal models of PD. ***Results:*** In this work, our results showed clear modifications in the expression of kinases involved in mTOR and PKR apoptosis pathways, in lymphocytes of PD patients treated or not with anti-PD treatment (levodopa), which confirmed the role played by apoptosis in the pathogenesis of this disease and the positive effect of treatment with medication on this parameter. Others proteins involved in apoptosis were also evaluated in lymphocytes of patients as the expression of the peripheral benzodiazepine receptor and caspase-3. ***Conclusion:*** Translational control is altered in PD and hence its evaluation in peripheral blood mononuclear cells may serve as an early marker of apoptosis and indicate the efficacy of the dopaminergic treatment.

Introduction

Parkinson disease (PD) is characterized by a progressive loss of dopaminergic neurons of the substantia nigra pars compacta projecting to the corpus striatum, which triggers a cascade of profound modifications in the functional organization of the basal ganglia circuitry, leading to the expression of PD motor symptoms that are significantly improved by dopamine [1]. It has been proposed that, in PD, defective regulation of apoptosis is likely to play a key role in nigral cell death [2]. At the molecular level, cell death has been associated with an imbalance between the antiapoptotic mammalian target of rapamycin (mTOR) pathway and the proapoptotic double-stranded RNA-dependent protein kinase (PKR) pathway. The mTOR pathway leads to activation of protein synthesis by phosphorylation of the kinase p70S6K. This pathway also regulates cell growth and proliferation and can attenuate

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karger@karger.com
www.karger.com/ndd

Dr. Stéphanie Pain
Faculté de Médecine-Pharmacie, Université de Poitiers
6 rue de la Milétrie, BP 199
FR–86034 Poitiers Cedex (France)
stephanie.pain@univ-poitiers.fr

apoptosis. The PKR pathway, which decreases proteins synthesis, is associated with cellular apoptosis [3–5]. This fact is further support by studies showing an increase of PKR phosphorylation in the hippocampus of PD patients [6] and in the brain of an animal experimental model of PD, a reduced activity of mTOR along with a significant activation of the proapoptotic PKR pathway [7–9].

Interestingly, the precise interactions between these molecular modifications and dopamine replacement therapy, which has been experimentally suggested to be neuroprotective [10, 11], remains unknown.

Due to the relative inaccessibility of the central nervous system, mechanisms potentially relevant to PD pathogenesis have been studied, in PD patients, using other cell systems. In particular, peripheral blood cells, such as lymphocytes and platelets, have been used to demonstrate that some of the changes observed at the nigral level (decreased activity of mitochondrial respiratory chain or increased oxidative stress) can also be observed peripherally [12–14]. Interestingly, some authors showed a decrease in mTOR expression in lymphocytes of PD patients with postural instability [15].

In the current study, we investigated, in peripheral lymphocytes, the activation of pro- and antiapoptotic pathways in lymphocytes of naive PD patients (i.e., de novo) and PD patients treated by dopamine replacement therapy as compared to a control patients group. The aim of our study was to determine the lymphocyte expression of proteins involved in apoptosis regulation, such as kinases activated in protein synthesis pathways. We also evaluated the expression of the peripheral benzodiazepine receptor (PBR), which participates in a key step of the apoptotic process (such as the opening of the mitochondrial permeability transition pore) [16, 17], and the activation of caspase-3, which is a major effector of the apoptotic cascade.

Patients and Methods

Forty-four PD patients were examined at the Department of Neurology of Poitiers University Hospital (CHU Poitiers), France and included in our study. The control group consisted of 19 healthy blood donors (14 female, 5 male; age 59.9 ± 10.3 years) matched to the PD group with regard to age and education level. Fifteen de novo diagnosed PD patients (3 female, 12 male; age 62 ± 9.5 years) were tested before the onset of treatment and were considered for the study only after confirmation of the diagnosis. The mean duration of PD before the inclusion was 19.1 months. Finally, 27 PD patients (7 female, 20 male; age 60.9 ± 8.5 years) were under anti-PD treatment (levodopa). The mean duration of PD before the inclusion of these subjects was 130.3 months. Blood samples (10 mL) were used for biological analyses.

Preparation of Lymphocytes

Mononuclear cells were isolated from blood samples using Ficoll-Histopaque 1077 density gradient centrifugation at 700 *g* at 4 °C for 20 min. The lymphocytes were then washed in phosphate-buffered saline (154 mM NaCl, 1.543 mM KH_2PO_4, 2.7 mM Na_2HPO_4; Sigma, France) and resuspended in 200 µL of lysis buffer (50 mM Tris-HCl, 50 mM NaCl, pH 6.8, 1% Triton X-100, 50 mM NaF, 1 mM PMSF, protease inhibitor cocktail [1 µL/1.10^6 cells], 1% [v/v] phosphatase inhibitor cocktail). After homogenization, lysates were sonicated for 2 min and centrifuged at 15,000 *g* for 15 min at 4 °C. The supernatant was saved and analyzed for protein determination using a protein assay kit (BioRad, France). Aliquots were frozen at –80 °C until further analysis.

Western Blots

The levels of the protein kinases mTOR, PKR, and p70S6K were measured by the ratio phosphorylated form/total form and analyzed by Western blotting. Specific sites were screened with antibodies for mTOR (Ser 2448), PKR (Thr 446), p70S6K (Thr 389), and cleaved caspase-3 (asp 175) (Cell Signaling Ozyme, France) and for PBR (Trevigen, Interchim, France). Twenty micrograms of protein (per sample) were separated on either 3–8% Tris acetate gels (mTOR and p70S6K) or 4–12% Bis-Tris gels (PKR, PBR, and caspase-3) and transferred to polyvinylidene difluoride membranes (Fisher, France). The membranes were washed twice in Tris-buffered saline Tween-20 (20 mM Tris-HCl, 150 mM NaCl, pH 7.5, 0.05% Tween 20) and incubated in 5% nonfat milk containing 0.21% sodium fluoride with primary antibody in blocking buffer at 4 °C. After 48 h of incubation, membranes were washed twice with Tris-buffered saline Tween-20 and then incubated with the secondary antibody peroxidase-conjugated antirabbit IgG (1:1,000 dilution; Amersham Biosciences, France) during 1 h at 25 °C. Finally, membranes were washed again and exposed to ECL Plus (Amersham Biosciences). Chemiluminescence was captured by the G:Box real-time imaging system (Syngene). The data for each protein were compared to the data of the corresponding actin (mouse monoclonal anti-β-actin antibody, 1:1,000; Sigma). Then, the ratio of phosphorylated protein/total protein for mTOR, PKR, and p70S6K was calculated to determine the level of protein activation. All protein levels were adjusted to an internal control obtained from neuroblastoma cells in order to standardize Western blot assessments.

Statistical Analysis

Results are expressed as means and SEM. The level of significance was $p < 0.05$. Comparisons between groups were done using Scheffé's test after a Kruskal-Wallis ANOVA.

Results

The level of the phosphorylated protein kinase mTOR was significantly reduced in lymphocytes of de novo PD patients (46%, $p < 0.01$) and of treated PD patients (57%, $p < 0.01$) compared with controls (Fig. 1a), but no significant difference was observed in both PD patient groups. In the same antiapoptotic pathway, no significant alteration of p70S6K activity was demonstrated in de

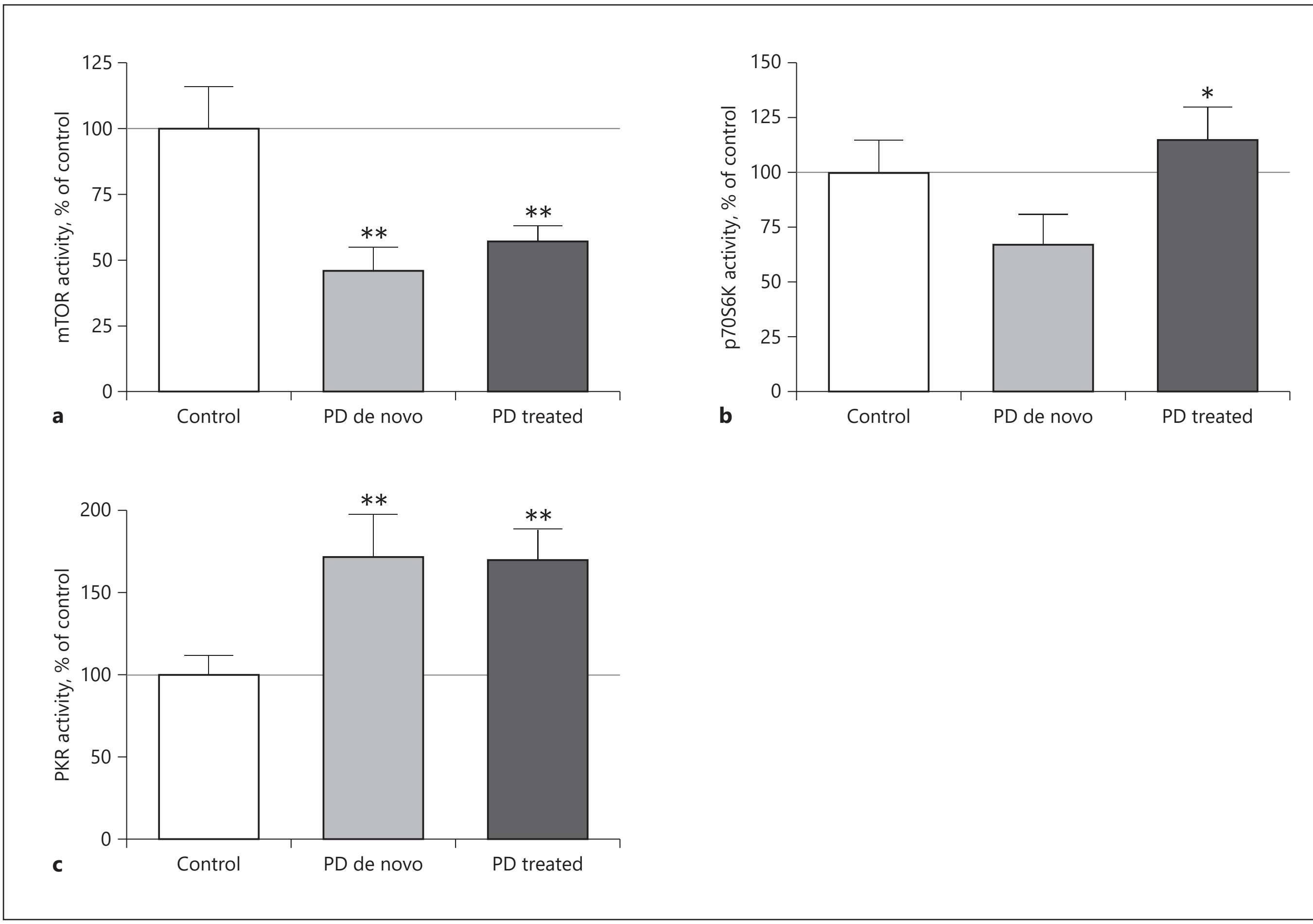

Fig. 1. mTOR, p70S6K, and PKR activity in lymphocytes of PD patients. Activation of mTOR, p70S6K, and PKR, calculated by the ratio between the phosphorylated form and the total form of each protein, is expressed as percentage of control subjects and is shown as mean ± SEM. **a** Results of mTOR activity in control, de novo PD, and treated PD patient lymphocytes. ** $p < 0.01$ versus control patients (ANOVA Kruskal-Wallis test followed by Scheffé post hoc test). **b** Histogram of p70S6K activity in lymphocytes of control, de novo PD, and treated PD patients. * $p < 0.05$ versus treated PD patients (ANOVA Kruskal-Wallis test followed by Scheffé post hoc test). **c** Results of PKR activity in control, de novo PD, and treated PD patients. ** $p < 0.01$ versus control patients (ANOVA Kruskal-Wallis test followed by Scheffé post hoc test). mTOR, mammalian target of rapamycin; PD, Parkinson disease; PKR, double-stranded RNA-dependent protein kinase.

novo PD patients and treated PD patients versus controls ($p = 0.09$ and $p = 0.34$, respectively; Fig. 1b). However, p70S6K activity was found to be significantly increased in treated PD patients as compared to de novo PD patients (+48%, $p < 0.05$). The level of the phosphorylated PKR was significantly increased in lymphocytes of de novo PD patients (172%, $p < 0.01$) and treated PD subjects (170%, $p < 0.01$) compared with controls (Fig. 1c), but the values obtained for treated or de novo PD patients were not significantly different.

For the final apoptotic markers, the results in Figure 2a indicated a high PBR expression in lymphocytes of de novo PD patients (222.2%, $p < 0.05$) compared to controls. For dopamine-treated subjects, the PBR level in lymphocytes returned to basal value since no significant difference was observed related to controls. Finally, the activated caspase-3 level shown in Figure 2b was increased in lymphocytes of de novo PD patients (207.3%, $p < 0.05$) compared to controls. Dopaminergic treatment partially blocked caspase-3 activity since no significant difference of the cleaved caspase-3 level was observed in lymphocytes of treated patients compared to control and de novo PD subjects.

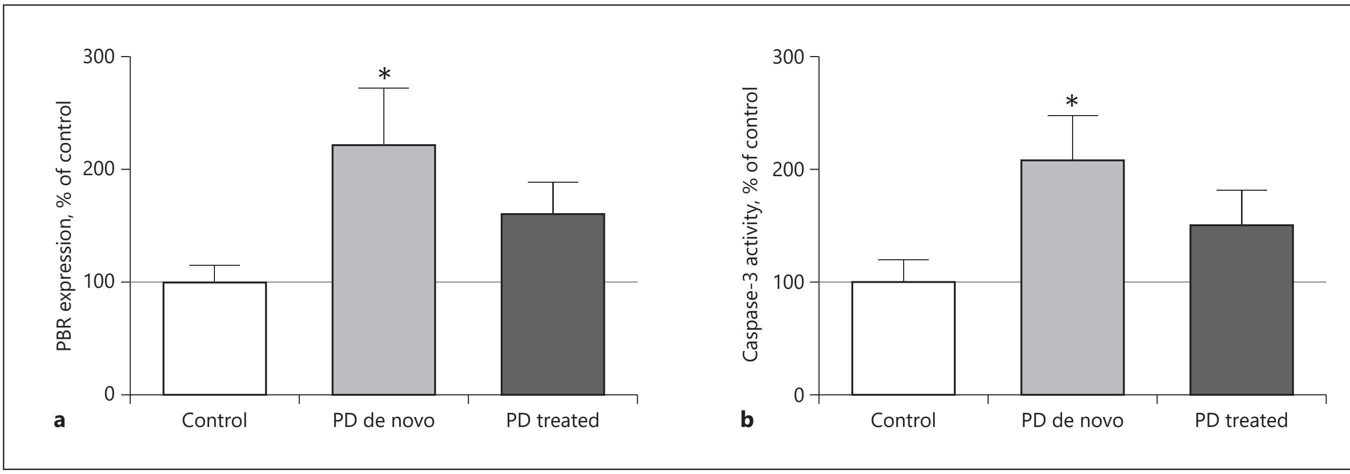

Fig. 2. PBR expression (**a**) and caspase-3 activity (**b**) in lymphocytes of PD patients. **a** The expression of PBR is shown as percentage of control subjects and is the mean ± SEM calculated in lymphocytes of control, de novo PD, and treated PD patients. * $p < 0.05$ versus control patients (ANOVA Kruskal-Wallis test followed by Scheffé post hoc test). **b** Activation of caspase-3 measured by the expression of cleaved caspase-3 is expressed as percentage of control subjects and is the mean ± SEM calculated in control, de novo PD, and treated PD patients. * $p < 0.05$ versus control patients (ANOVA Kruskal-Wallis test followed by Scheffé post hoc test). PBR, peripheral benzodiazepine receptor; PD, Parkinson disease.

Discussion

Defective regulation in the control of apoptosis may play a key role in the pathogenic mechanisms of PD. Apoptotic cell death and increased levels of proapoptotic molecules are present in the substantia nigra of PD patients [18]. Concerning human in vivo studies, because of the inherent difficulty of studying PD pathogenic mechanisms in the central nervous system of patients, peripheral blood cells, particularly lymphocytes, have been used as an accessible source of information on biomolecular aspects relevant to PD pathogenesis (such as changes in superoxide dismutase activity [19] and DNA damage) that point to the existence of enhanced oxidative stress [20].

The aim of our study was to determine the lymphocyte expression of proteins involved in apoptosis regulation, such as kinases activated in pathways of protein synthesis. We also evaluated the expression of the PBR, which participates in a key step of the apoptotic process, and the activation of caspase-3.

In this study, we sought to investigate whether modifications of proteins involved in apoptosis in PD can be detected outside the central nervous system (i.e., in peripheral blood lymphocytes) and how pharmacological dopaminergic treatment interferes in this regulation. Indeed, there is currently no agreement on the effect of anti-PD therapy on the mechanisms of cell death [21].

The main finding of our study is the alteration of the expression of kinases involved in protein synthesis pathways along with an increased activity of caspase-3 and elevated PBR expression observed in lymphocytes of de novo PD patients compared to healthy controls.

The current study is the first to investigate mTOR and PKR pathways in peripheral cells of PD patients. Indeed, the levels of mTOR and p70S6K, which are kinases with mainly antiapoptotic properties, are decreased, while the proapoptotic pathway involving PKR is increased in de novo PD lymphocytes compared with those of age-matched controls. The fact that such alteration is also present in nonneuronal cells, such as lymphocytes, is consistent with the hypothesis that in PD patients, the central molecular proapoptotic process reflecting their pathology could also be observed in lymphocytes, and that these cells should be considered as a biomarker for PD [22]. However, the latter hypothesis is challenged by the fact that some changes in mTOR and PKR pathways were also demonstrated by our research group in lymphocytes of Alzheimer patients [23, 24]. This suggests that the molecular alterations reported here might represent a nonspecific marker of degeneration. Indeed, these modifications in the levels of proteins involved in the control of apoptosis in lymphocytes of PD patients have also been shown for caspase-3 [25, 26], superoxide dismutase activity [19], and Akt activity [27]. Interestingly enough, these changes were modulated by the pharmacological dopaminergic treatment of PD patients [25–28].

In the current study, pharmacological treatment seemed to partially modulate changes observed in lymphocytes of PD patients. The influence of pharmacological treatment with dopaminergic agents in intracellular biochemical pathways, particularly those involved in the cell death process, has long been debated [29]. While dopaminergic treatment did not change mTOR and PKR activity in lymphocytes of PD patients, dopamine-treated patients showed p70S6K levels which returned to the protein expression observed in control subjects. Moreover, untreated PD patients showed increased PBR levels and caspase-3 activation compared to control subjects, values which tended to decrease in patients treated with levodopa. Nevertheless, further data based on a larger cohort of patients will be necessary to confirm these findings.

Some studies have already showed that dopaminergic stimulation is associated with complex changes in the intracellular levels of apoptosis-regulatory proteins (decrease in caspase-3 activity), and this may play a role in the therapeutic response to dopaminergic agents [25, 26, 28].

In conclusion, we found, for the first time, clear modifications in the expression of kinases involved in the mTOR and PKR apoptosis pathways in lymphocytes of PD patients, which confirms the role played by apoptosis in the pathogenesis of this disease and clearly shows that these kinases are not specific to a neurodegenerative disease. Moreover, our results confirming the impact of dopaminergic treatment on apoptotic processes in the peripheral blood of PD patients supplement the understanding of the positive effect of dopaminergic medication in PD patients.

Statement of Ethics

The regional ethics committee of the region Poitou-Charentes (France) approved the study, and patients were asked to sign an informed consent form consistent with institutional guidelines.

Disclosure Statement

The authors have no conflicts of interest to declare.

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Author Contributions

All authors made significant contributions and approved to this submitted manuscript. The introduction, experimental measures, and literature review were written by S. Pain, J. Deguil, and P. Barraud. J. Belin and J.-L. Houeto were clinicians who recruited the patients. S. Ragot realized the statistical tests. All authors contributed to the discussion.

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